

Design and implementation of an online bookstore: The case of University of Ilorin (UNILORIN) Bookshop

Olubukola Oluyemisi Ajiboye

Department of Library and Information Science, University of Ilorin, Nigeria

E-mail: getbukkyajayi@gmail.com

Mufutau Ayobami Asiru

Principal Librarian

Federal Polytechnic, Ado-Ekiti

E-mail: ayobamimufu@gmail.com

Taiye Emmanuel Ogunlade

E-mail: ogunlade.emma@gmail.com

Abstract

This study aimed at developing an online platform where books and other items are made available online for convenience, faster way of checking product details and making purchase online at University of Ilorin (UNILORIN) Bookshop. This addressed the problem of searching for availability and comparing prices of textbooks physically which can be exhausting and time consuming in one's search for information. The research methodology adopted the use of questionnaire and observation to determine bookshop users' needs of an online bookstore. The results showed that 80% of the respondents indicated interest in buying books online. Observation was used to gather information about the existing system's operations particularly on price update of books and collection of new books. The inadequacies of the existing system were improved on by the proposed system. The new system made use of a markup language HTML, PHP (Hypertext Preprocessor; a server-side scripting language) as the main code for the program, CSS framework Bootstrap, CSS (Cascading Style Sheet) MySQL as the database for the system. This project has been able to browse through product categories, use the search option, check product description and price, add to cart and also increase the number of Item, select the delivery method and finally, make payment. It was recommended that further work be done using another web stack such as Nodejs, MongoDB, Ruby on Rails or design a mobile app using hybrid technology.

Keywords: Design, implementation, online, bookstore, product

Introduction

Electronic commerce commonly known as e-Commerce which is a subset of e-Business consists of buying and selling of products or services over electronic systems such as the internet and other computer networks. E-commerce is the use of the Internet for marketing, identification, payment and delivery of goods and services. Through the e-commerce technology, the Internet has revolutionized the mode of business transactions by providing consumers with the ability to bank, invest, purchase, distribute, communicate, explore, and research from virtually anywhere, anytime

where there is Internet access Anup (as cited in Ayo 2011).

A user who accesses the website is able to select the desired product that he/she wants to buy which is added to their cart (list of intended online purchases a customer want(s) to acquire, after which the customer proceeds to fill a billing address, shipping address, a delivery option and payment information such as credit card details.

Despite the convenience of online shopping, not everyone chooses to purchase items and services online. Some people like the idea of physically going to a store and experiencing the shopping process. They like to touch the merchandise, try on

clothing, and be around other people. Online shopping doesn't permit shoppers to touch products or have any social interaction. It also doesn't allow them to take the merchandise home the same day they buy it.

Online shopping also eliminates the catchy, yet irritating music, as well as the hundreds, if not thousands, of other like-minded individuals who seem to have decided to shop on the same day. Online shopping allows you to browse through endless possibilities, and even offers merchandise that's unavailable in stores. If you're searching for a niche product that may not be distributed locally, you're sure to find what you're looking for on the internet. What's even more useful is the ability to compare items, similar or not, online. You can search through multiple stores at the same time, comparing material quality, sizes and pricing simultaneously.

Existence of online platforms to facilitate on campus book selling has increased throughout the years as the rise of agile information sharing and retrieval become critical factor in processes that occurs within campus proxy. Information dissemination which occurs traditionally have resulted in knowledge obsolescence as in today's world knowledge is rapidly changing as new innovation comes in. Over the years, many Universities have embraced the adoption of having an online bookstore, some of the Universities Are University of Oregon, University of Washington, University of Colorado, University of California Los Angeles and many others.

In order to successfully nurture creativity and innovativeness among its students, Universities must strive to focus on aspects which relate to providing quality academic learning, which starts from dissemination and utilization of literature. Examples of online bookstore in Nigeria are

Lantern books, iQrbooks, and sunshine bookseller

Statement of the problem

In an academic environment, pivotal to learning are books and academic resources that aid the students to attain their academic goals and academic staff to facilitate teaching and research. It becomes expedient that there should be access to current and relevant academic books and learning materials to keep abreast with the trends in the various disciplines. The responsibility to purchase these books at the bookstores rests on both the staff and students. To make purchase, they must check for its availability, ask for the price, and compare prices in order to make an informed decision. Unfortunately, some end up walking from a long distance to the bookstore to find out that the item they wish to buy is not available, or the price is more than the price assumed by the potential buyer, some spend more time in searching for a book. All these lead to physical stress, which can be time consuming thereby leading to reduced productivity.

In order to ameliorate the problems highlighted above, it becomes pertinent to have an online platform where a client can search for, compare products, check the price, add to cart and fill the billing and payment addresses and card details to purchase items listed in their carts.

Objectives of the study

The main objective of this study is to design a responsive and efficient web application that will transform UNILORIN Bookstore from its traditional way of purchasing books to using an online platform which operates electronically, through the design and implementation of an online bookstore.

The specific objectives are to:

1. design an electronic commerce system,
2. develop this system using server side programming language, and
3. implement a payment system and create a method of delivery option.

Significance of the study

This system will assist patrons of the bookstore to access books and other items online in an effective manner. This project aims to solve problems associated to traditional way of purchasing books from the bookstore. The system gives convenience, quick access, and easy access to available items in the bookstore.

Scope and limitation of the study

This research work is centred on developing a system that provides an effective and efficient way of buying books and other academic materials in UNILORIN Bookstore. This study is limited to UNILORIN Bookstore, University of Ilorin.

Literature review

The origin of online retailing began in 1995 with the launch of eBay and Amazon. EBay was started in the 1995 by Pierre Omidyar as an online auction. It expanded to involve an online retail center where goods were purchased. Jeff Bezos established Amazon, which specialized itself in book sales. Amazon's website shows the company generated more than \$230 billion in revenue in 2018 while posting more than \$10 billion in profit or net income. Other e-retailers that operate exclusively online and compete with Amazon include Overstock.com and JD.com. Today, internet stands out to be the biggest medium for selling and buying.

Online retail has been one of the most upcoming retail sales in the genre of book sales. In the last decade, bookstores

have experienced a dramatic decline in their volume of sales, with the online retail marketing having an upsurge. The vast majority of the online retail websites are middlemen between the actual publishers and the consumer, with commissions being paid on each product sold. Consumers can compare the price, quality and availability of items from across the globe (Limayem et al., 2000). The power to displace the traditional markets lies in the ability to satisfy its consumers' needs better than those traditional business approaches and provide consumers with lower prices, wider selections, better choices, superior services, or more convenience. Vassos (as cited in Yang, Zielinski & Cheng, 2009).

Chevalier and Goolsbee (2003) examined the college textbook market to determine whether its consumers are forward-looking, whereby a forward-looking consumer is defined as a person who purchases durable goods today with the future in mind.

According to Muzumdar (2012), Low price, price flexibility and convenience stands out to be the biggest advantage online medium has compared to on & off campus bookstore. Online medium shows more interaction between buyer and seller, and also more flexibility in pricing policy.

The advantage of online price comparison is important as it enables and guarantees purchase at the lowest price possible after comparison with different sellers at specific time frames in the online market. In contrast to the traditional bookstore, the prices are fixed and it is difficult to move between bookstores to compare prices of books before making a decision.

In comparison to the traditional markets, books are almost always available in the online markets and globally networked information Systems can reduce

the search and transaction costs for buyers and sellers to locate and trade and can thereby facilitate the creation of technology-mediated electronic exchanges. This corroborates the findings from Chen et al (2012) which states that print textbooks in colleges are so expensive that the prices of two textbooks would be equal to an e-book-reader. Hence, with the super development of technology, electronic devices' prices are decreasing compared to a few years ago.

According to Malone, Yates and Benjamin (1987), comparing on and off campus bookstore shows that an online retail doesn't require an inventory to stock the sales, as every individual retailer, seller can sell his product through his individual inventory. Having multiple sellers helps online retail websites to keep up with the availability of the book throughout the year. One of the biggest benefits of online shopping is that you can buy almost anything you could imagine without ever leaving your house. Online stores are open 24 hours a day and are accessible from any location with an Internet connection. In general, online stores are able to carry more selection than traditional brick-and-mortar stores. Because online stores don't need to attractively display their items on shelves, they can keep a larger amount of inventory on hand. They also might only have small amounts of each item, since they don't need to display them, and can order more from their supplier as needed. Also, online shops tend to provide more information about items for sale than you would get in a physical store. Product descriptions most often include a description from the manufacturer, another description from the vendor, specific technical and size details, reviews from professional magazines and journals, and reviews from people who have bought the product. Online book stores often will have excerpts of the books (usually the

first chapter) for you to read. Having all this information available when you are considering a purchase makes you a more informed consumer without having to perform extra research yourself.

The ability to sell from anywhere by the traditional on-campus and off-campus market is important and exchange of money can also be done online for any item purchased. One generally has several options of payment methods when purchasing items online. The most common way to pay online is with a credit card; one almost never wants to pay for items online with cash or cash-like methods (e.g., debit cards, money orders, and personal cheques). Credit cards offer some additional protections from fraud, including zero-liability policies (one should not be held responsible for purchases made with one's stolen card) and credit protector services alert one to fishy activity. Importantly, credit cards are not directly linked to bank accounts, so no money will be withdrawn from account until monthly bills are paid. This accounts to a large extent for the success of on campus and off campus bookstores. To counter these problems, Amazon and eBay in last five years have started using direct checkout methods as Alternative to sign up accounts, where direct check out method doesn't require a person to feed in and save their account and credit card information (Yang, Zielinski and Cheng, 2009). Also third party online payment and money transfer websites like PayPal act as third party firms for money transfers. PayPal started in 1998 and currently acquired by eBay runs as the most trusted third party money transfer service around the globe. All eBay purchases, for example, are conducted via PayPal. PayPal and other payment websites will never ask for your credit card number or other personal information by e-mail.

In summary, online retail sales are preferred by the people who are more concerned about price difference in goods. Price is perceived to be one of the factors that is highly influential between both seller and consumer. Consumers like to exploit low prices, while sellers prefer to have price flexibility. Price comparison seems to be the most beneficial tool that the online retailing website has over other media. For consumer search engines, the function of “add to cart” and “check out as guest” are part of the advantages, but for sellers it is direct transaction with consumer, online money transfers, bundling products.

Nigeria is the fastest growing telecommunication country in Africa (Ayo et al., 2007). According to the Nigerian Communications Commission website, internet users marginally increased to 115,931,255 in March, 2019 ranking Nigeria at the top in the list of African countries with the highest number of internet users. Compared to January 2018, there has been a 4 million increase in the number of internet users. Over 112 million Nigerians had access to the internet in 2018, representing 56% of the population. This growth percentage represents the increase in the number of internet users between the years 2000 and 2019. In 2009, the figure went above double as 23,982,000 million people used Internet in Nigeria. By June 2010, the number of internet users in Nigeria has grown to 43,982,200 that is 29.5% of the country's population. The increasing users of internet in Nigeria from 0.1% in 2000 to 29.5% of its population in June 2010, revealed that the use of internet in the country is growing at a sporadic rate and still has the potential to grow higher. Nigeria's e-commerce market was recently valued at 225 billion Naira due to the 25% in e-commerce growth. The reason for this growth is not far-fetched as Nigeria is the

most populous nation in Africa. Despite the growth of internet users in Nigeria, much research work has not been done in accessing the B2C e-commerce activity. Presently, many online shopping sites are thriving in Nigeria, constantly thinking of how to satisfy customers by designing products aimed at servicing thousands of searchers every week. Some of them are: Jumia.com.ng, Konga.com.ng, Yudala.com.ng, Kaymu.com.ng, www.234world.com, Xtaples.net, www.book.sng.com, www.orderbay.com. Some of these sites make the transaction process very user-friendly which tends to make buyers prefer the online market. A site like 234world.com allows buyer to pay to a designated bank account after making online purchases. The items purchased are then shipped to the buyer at the speed of light. Soft Pay allows individual to receive money online and thereafter use it to pay for purchases made. It even helps online shopping sites to collect payment for purchases made.

Considering all these, it is expected that the number of people engaging in e-commerce activity will increase. E-commerce has however not been widely tapped into in Nigeria. Many Nigerians still treat its benefits with deep skepticism due to lack of trust. They do not believe that e-commerce transactions could be successfully conducted. In light of the difficulties facing the e-commerce sector in Nigeria, online vendors are devising techniques to encourage and increase customers' confidence in online shopping. Popular online vendors like Jumia and Konga have introduced the pay on delivery option where customers make their orders online without having to pay online; payments are made when the customer receives their goods. This has gone a long way with addressing the trust and security issues associated with buying products online (Gabriel et al 2016).

Methods

System development lifecycle

System development methodology is a standard set of steps used by organization to develop and support their information system, this often follows a lifecycle, the system development life cycle is a common methodology for systems development in many organizations, it marks the steps or phase in information system. Systems development methodology is a standard process followed in an organization to conduct all the steps necessary to analyze, design, implement, and maintain information systems.

System development lifecycle SDLC is divided into four stages namely Planning, Analysis, Design, Implementation and Operation

System planning is the first phase in the SDLC it has two primary activities which are; firstly, identify the need for a new system or enhanced system secondly; to investigate system and determine the proposed system's scope.

The second phase of the systems development life cycle is systems analysis. During this phase, the analyst thoroughly studies the organization's current system procedures and the information systems used to perform tasks such as general ledger, shipping, order entry, machine scheduling, and payroll. Analysis has several sub-

phases. The first sub-phase involves determining the requirements of the system to determine what the users want from a proposed system. it involves a careful study of any current systems, manual and computerized, that might be replaced or enhanced.

Questionnaire

Copies of questionnaire were administered randomly to students who came into the bookstore, questionnaire was used to get information about the need for an online bookstore. It contained both open ended and close ended questions

This method was used because questionnaire:

1. Can be analyzed more 'scientifically' and objectively than other forms of research
2. It gives the respondent freedom to express their view by using open ended question
3. A questionnaire is concise only relevant questions are asked.
4. Questionnaire is not subject to influence decision of the respondent. Fifty-seven (50) questionnaires were administered to patron/student. Result of the questionnaire are represented in charts below.

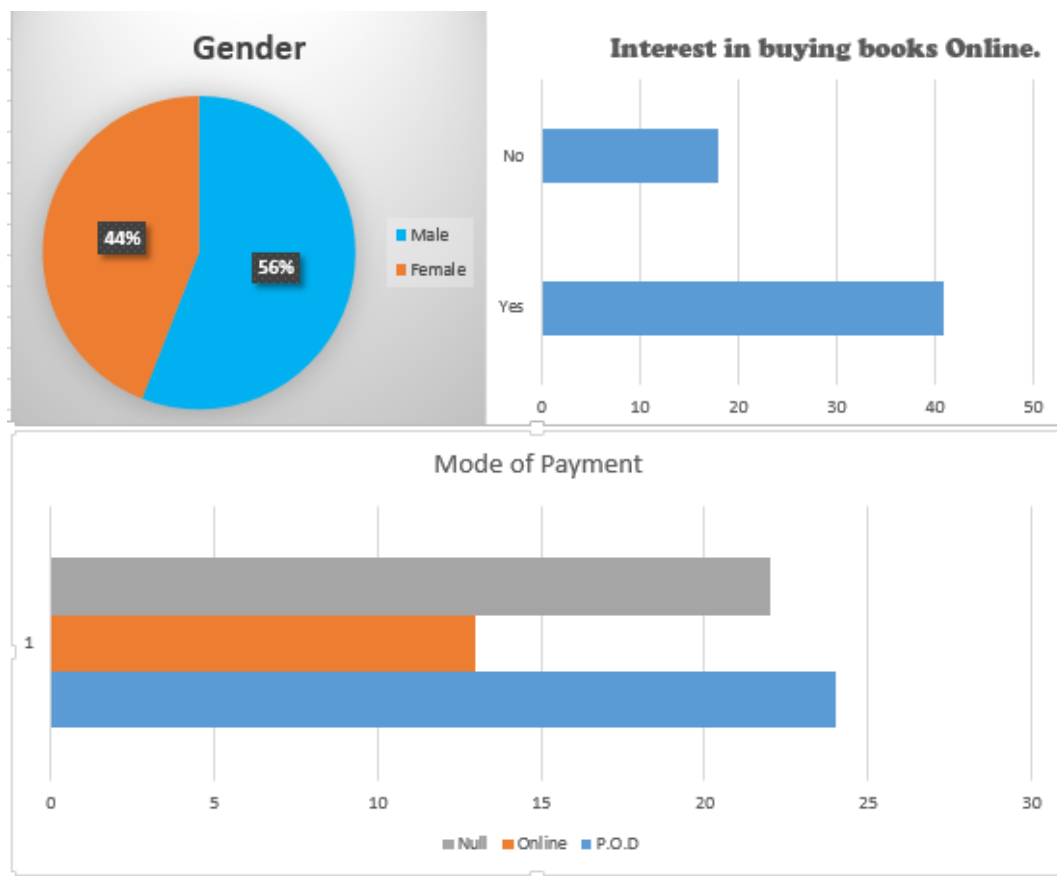


Figure 1: Gender, interest and mode of payment online

Observation

For the purpose of observation, in order to understand how the traditional system works, the researcher went to the bookstore on several occasion to carefully watch and record the way operations are being carried out. This technique involved gathering of information on how the existing system running of operations in the bookstore, particularly about the update on price of books and collection of new books. It is a

very tedious and time consuming process which has its merits such as:

1. It gives room for proper details which are not recorded on document or gotten through the questionnaire.
2. It gives more detailed to know how user behaves when navigating through the books in the bookstore.
3. It also gives room to know the categories of books in the bookstore and the rise of some books.

System analysis

During this phase, the analyst thoroughly studies the organization's current system procedures and the information systems used to perform tasks such as general ledger, shipping, order entry, machine scheduling, and payroll. Analysis has several sub-phases. The first sub-phase involves determining the requirements of the system to determine what the users want from a proposed system. It involves a careful study of any current systems, manual and computerized, that might be replaced or enhanced.

Description of the existing system

The current system in use by UNILORIN Bookstore is completely manually based. The attendant at the bookstore takes new product, records it, and writes the price on it before placing it on the shelves. Also when a patron of the bookstore purchases the product, the attendant is responsible for taking record of the price, quantity, and issuing of receipts and also recoding it on the book set aside for that purpose. Overtime, these records are piled up and also they are prone to physical hazard that can result to loss of data and information.

Problems of the existing system

The following are some of the problems faced with the existing system.

- The system does not have any form of backup in case of fire outbreak or any other disasters.
- It is prone to errors due to manual method of handling information which are noticed in processed information.
- It becomes a problem when many buyers come in to get product thereby leading to long queue in the bookstore
- This system is not available 24/7.

- This system does not give room for proper comparison of items in the bookstore.
- It is time consuming as a user will need to search through the shelves before finding the item or books desired.

Improvement of the proposed system over the existing system

The proposed system is concentrated on easing the management of records of items and other material by developing a bookstore in an online environment. The system would manage all important record about items, book and authors with the use of information technology tools. The creation of a web based application which can be accessed from anywhere in world in digital format of all existing records of the books and items in the bookstore. The system would make product categories, price authors details, and payment processing online available.

- Reducing the complexity of the system and maintain large information.
- Record will be efficiently maintained through database management system.
- Fast retrieval of documents through search using author, title of books.
- Convenience in making comparison between products
- Student can search for product 24/7
- Security of the web application will be ensured by putting SSL (Secured Socket Layer) and HTTPS in place
- The system provides easy search for records.
- Ability to perform adhoc query to the database by the administartor.
- There will be more data integrity.

- It eliminates the fear of people reporting criminal or hoodlums in their communities.
- Ability to make payment and print receipts online
- It eliminates queuing and helps make better decision

System architecture and design unified modelling language (UML)

This is a graphical language for visualizing, specifying and constructing the artifact of a software-intensive system. It offers a standard way to write standard system blueprints, including conceptual things such as business processes and system functions as well as concrete things such as programming language statement,

database schemas and reusable software components.

Architecture of the proposed system

The architecture supports the existing infrastructure requirement and provides specific recommendation for hardware, software and network structure based on the existing and projected user needs. The application requirements, data resources and people within an organization are all important in determining the optimum hardware solution. This is represented with three tier architecture that comprises of the User interface, Process management and Database Management System (DBMS). This gives a clear view of what the components of the system, the services they provide and the way these services communicate to bring about proper system functionality.

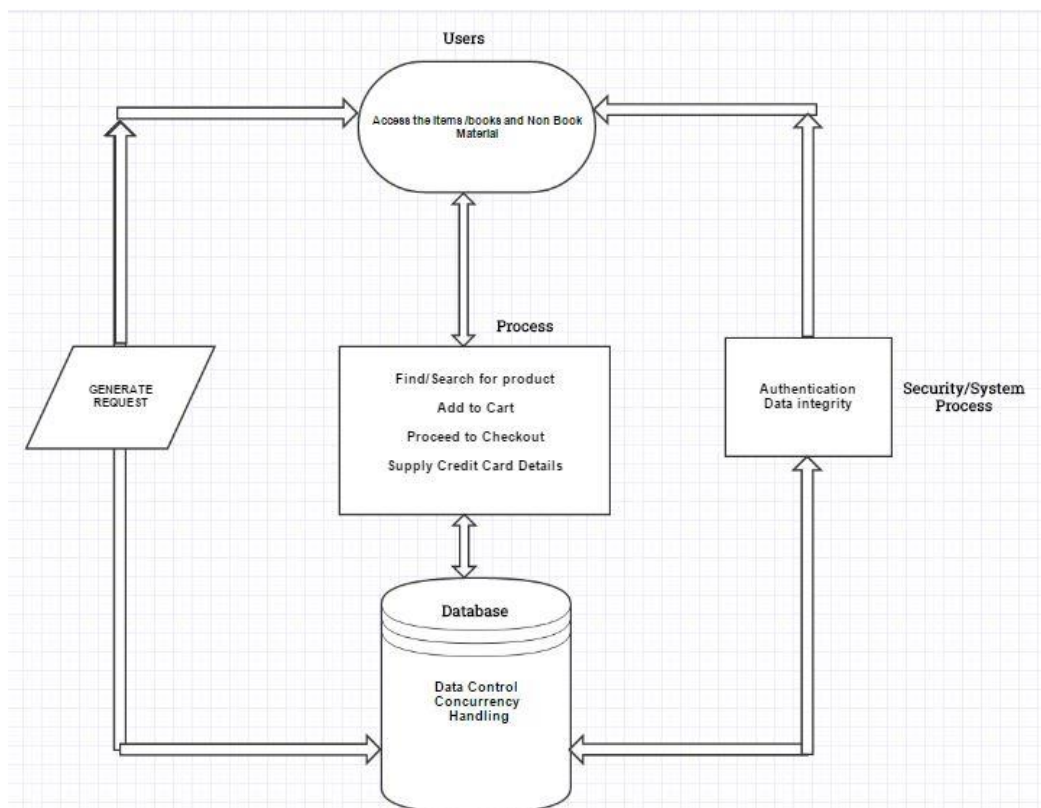
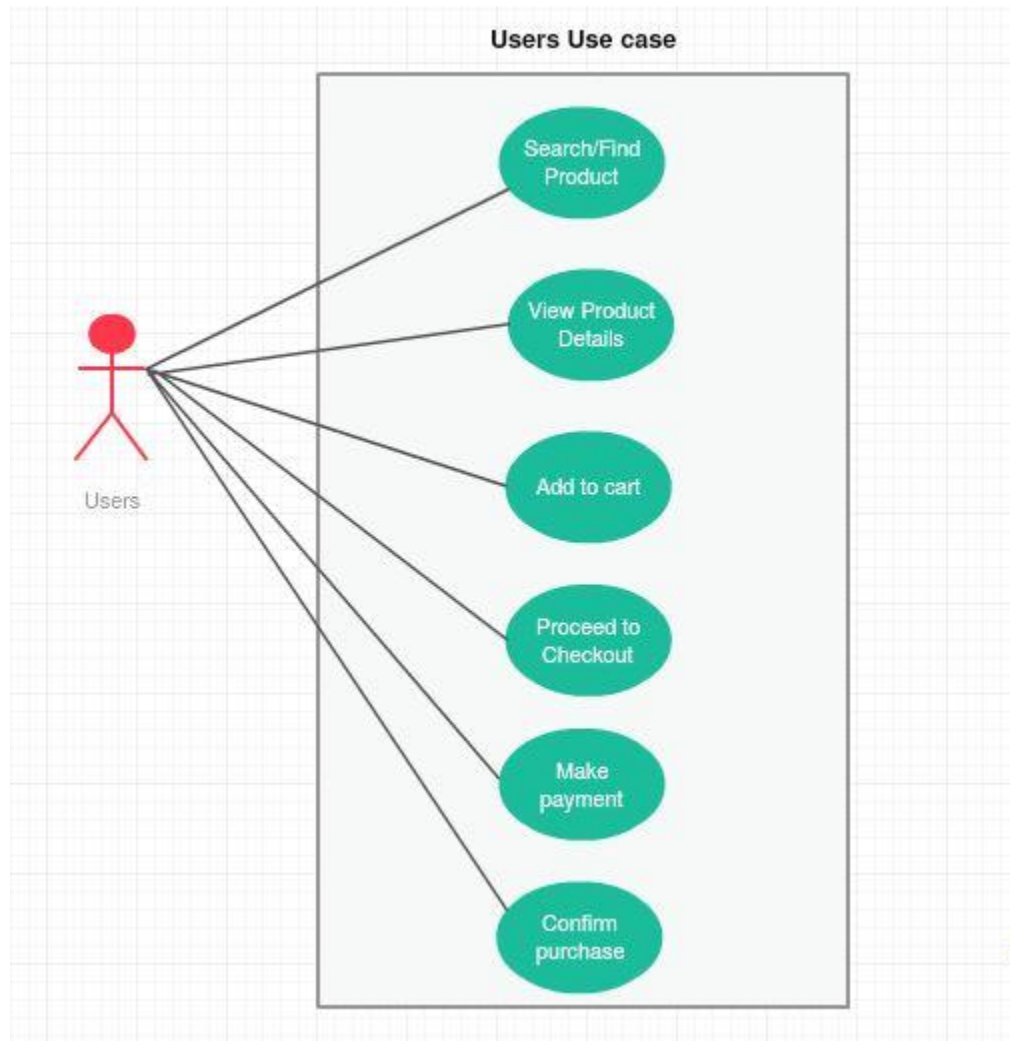


Figure 2: The architecture of the proposed system

Use Case

Use case is a list of step for defining interaction between a role and the system from the user perspective in the way they understand. Use cases are important requirement techniques that have been widely used in modern software engineering. A use case defines the

interactions between external actors and the system under consideration to accomplish a goal and the actor can a person or an organization. It is valuable for system developers; it is an effective technique for gathering systems from user point of view. Use cases focus on the behavior of the system from external point of view.



- **Figure 3: User use case diagram of the proposed system**

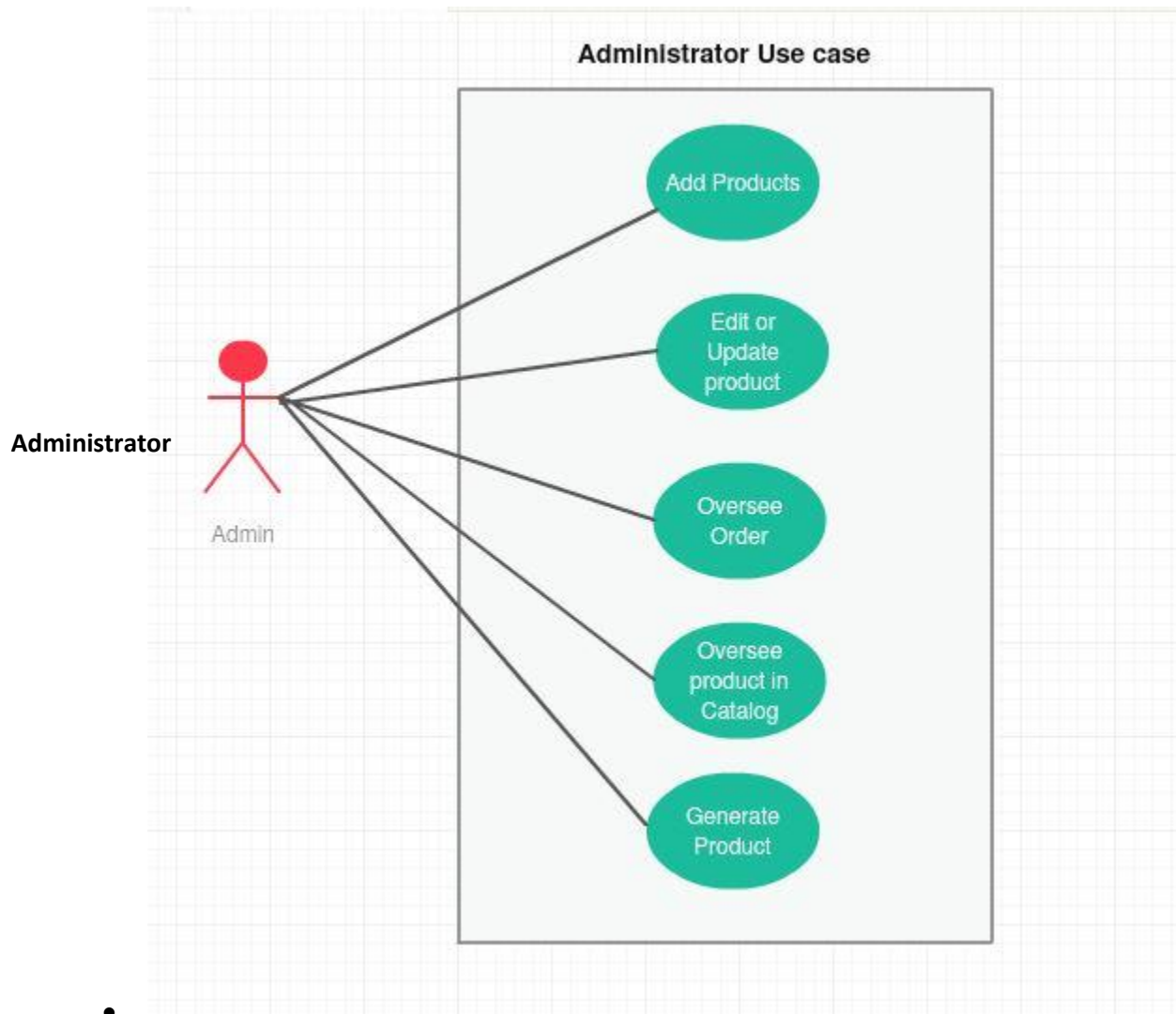


Figure 4: Administrators use case diagram of the proposed system

Flow chart

This is a diagram or graphic that represents a process or algorithm. The steps are represented by a series of boxes or other

specialized symbols with the description of each process and step connected with arrows.

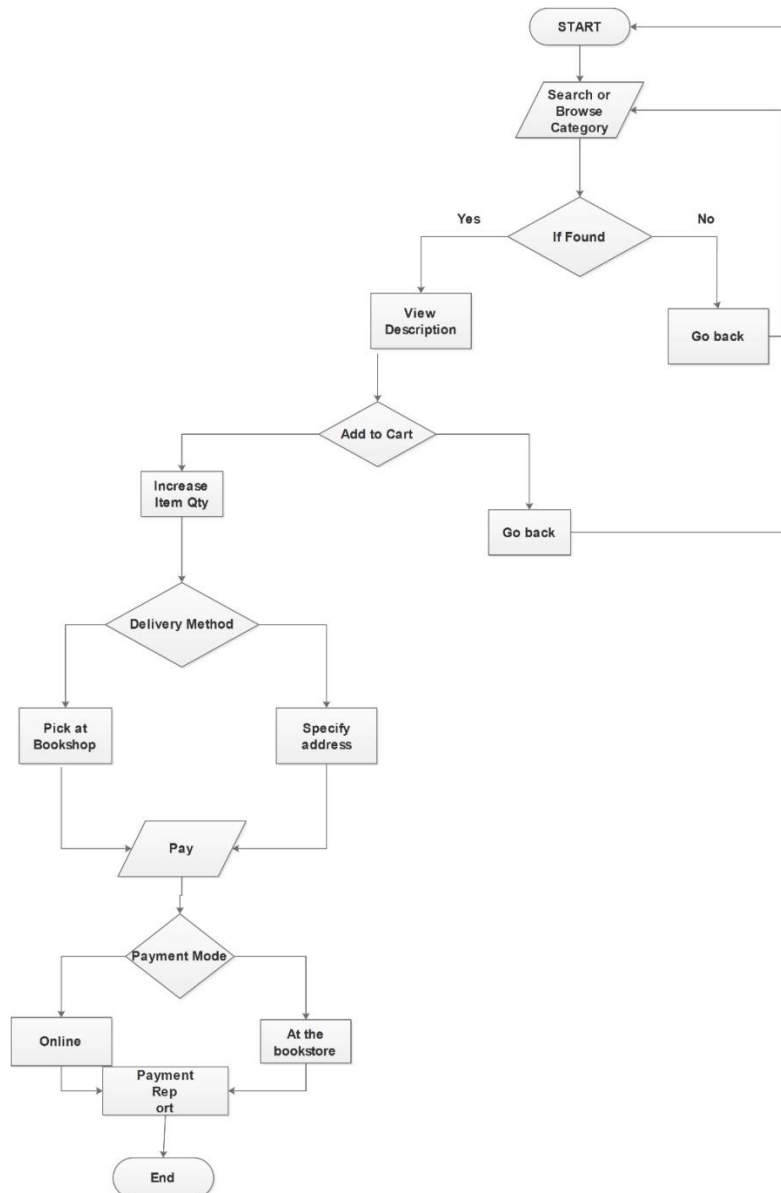


Figure 5: The flowchart online shopping

Justification of the development tools

These tools adopted for the designing of this system were used because of how well they

integrate with each other and proven for the development of similar systems.

Hypertext mark-up language 5 (HTML5)

HTML is a standard for creating website pages with elements consisting of *tags* enclosed in angle brackets such as <html>. HTML5 is a revision of the traditional HTML 4.01 and XHTML. HTML5 is a markup language used for the structure and presentation of World Wide Web contents and is based on tags that markup text-based documents. HTML5 was developed to solve compatibility problems that affect the current standard HTML 4.01 style syntax some include proprietary plugin and APIs (when a web page developed in one browser may not be able to load in another browser). The major advantage of HTML5 is the ability to support multimedia on mobile devices. HTML5 also included some new features which can really change the way users interact with documents including:

- New parsing rules for enhanced flexibility
- New attributes
- Elimination of outmoded or redundant attributes
- Drag and drop capabilities from one HTML5 document to another

Cascading style sheet

Cascading Style Sheet (CSS) is used to control formatting of element within the layout of website pages and it can be created within a HTML or within an external style sheet. CSS only functions with the existence of an html code. They can be used to define text style, color and size. CSS is used for creating styles that can be used repeatedly in a web page which takes less typing than the equivalent formatting with tags, thus saving you kilobytes of disk space. Apart from styling which CSS is generally known to be used for; the structure of applications can be built using CSS, meaning the application

would be table less. This attribute increases its demand in today's web designing.

Hypertext pre-processor (PHP)

PHP is a scripting language that is widely used by a lot of developers. PHP supports MySQL database and a variety of other database such as Sybase, Oracle and PostgreSQL. This flexible support is useful if you ever need to move your application to different database (Bacon, 2007). PHP is well optimized to with a fast response time on web applications. This can have a large base of core classes which can be used to build Graphical User Interface (GUI). PHP has the advantage of been scalable and a static language.

MySQL (Structured Query Language)

This is an open source and a comprehensive relational database server. It been used to develop web software applications. It supports large database with up to 50 million rows, it is fast and can run on different operating systems. MySQL offers exceptional security features that ensure absolute data protection. In terms of database authentication and mechanism for ensuring only authorized to have access to the database.

Javascript

JavaScript is an interpreted language with a C programming language like syntax. It is embedded directly into HTML page to make them responsive. JavaScript is a lightweight scripting language which make web page to be interactive. They are used to add more functionality, validate forms, create cookies and detect browsers.

Database Management System

Database management system is a software system that uses a standard method for cataloging, retrieving and running queries on data. The DBMS manages incoming data,

organizes data and provide ways for data to be modified or extracted by user or other programs (Tech Terms, 2014) other to

modify. DMBS stores data in such a way that is a lot easier to retrieve, manipulate and help in structure information.

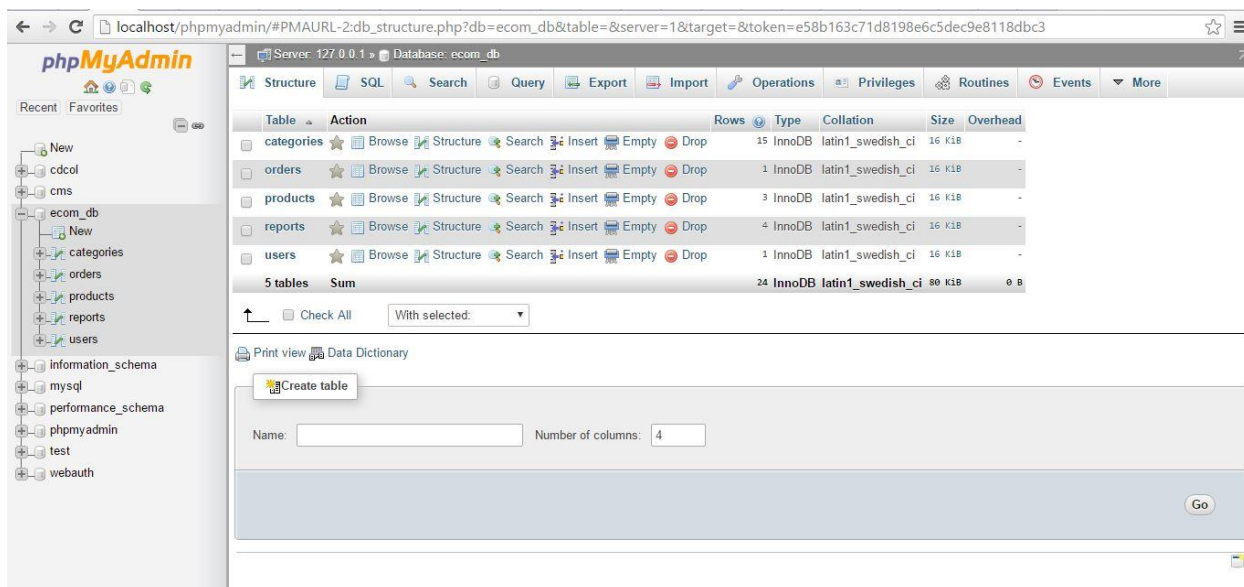


Figure 6: Screenshot of database table

Bootstrap framework

Bootstrap is a free and open-source collection of tools for creating websites and web applications. It contains HTML- and CSS-based design templates for typography, forms, buttons, navigation and other interface components, as well as optional JavaScript extensions. It uses 12 grid layout for it responsive design.

System implementation, testing and evaluation

Presentation of the system interface

This shows the system interface of the design environment of each window in the

E-Police Management system. This includes some of the user page, product, categories and Administrative page.

Homepage

This is the first page which loads when the application is started, it contains a slider with three slides at the center, categories section, navigation menu such as checkout page, and product list beneath the slider.

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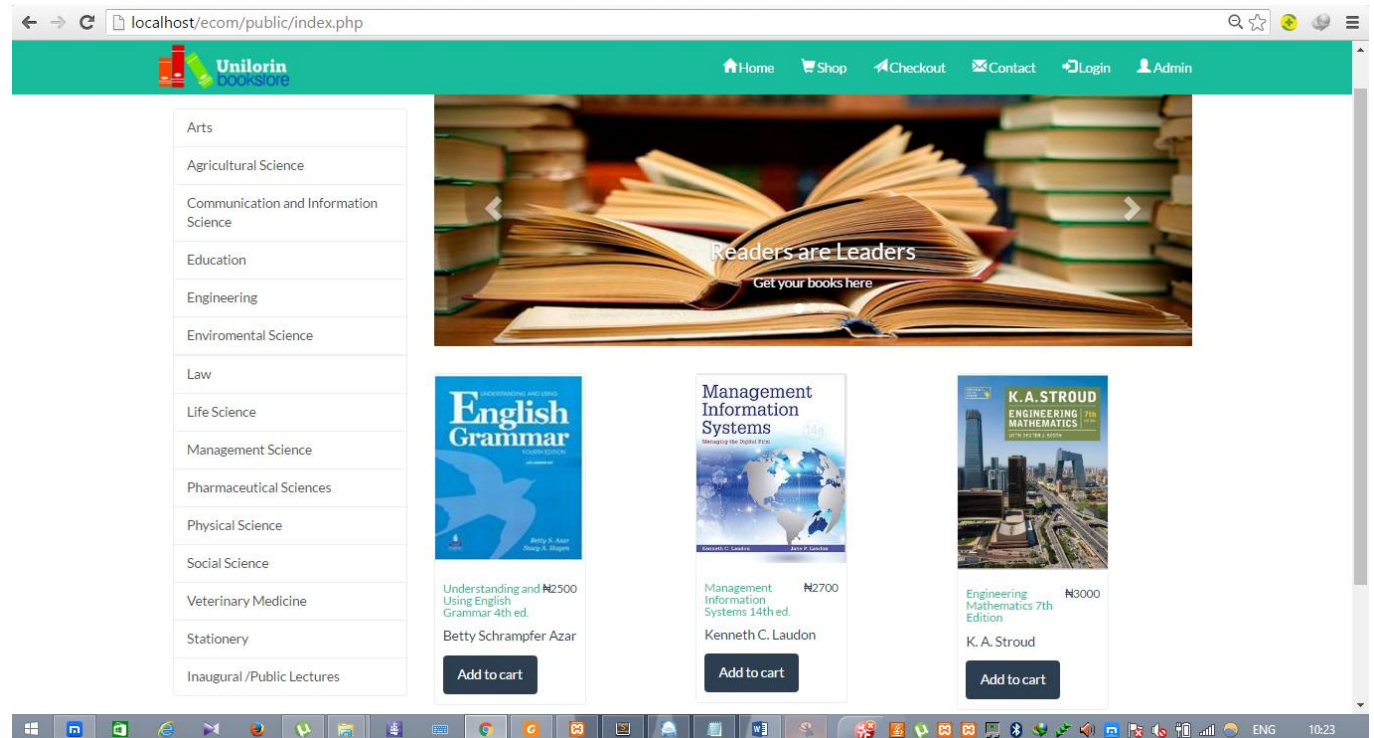


Figure 7: Homepage.

Administrator login page

This is the page which grants the admin authorization to admin page where the

admin can add product, modify product details, oversee reports, orders and delete product.

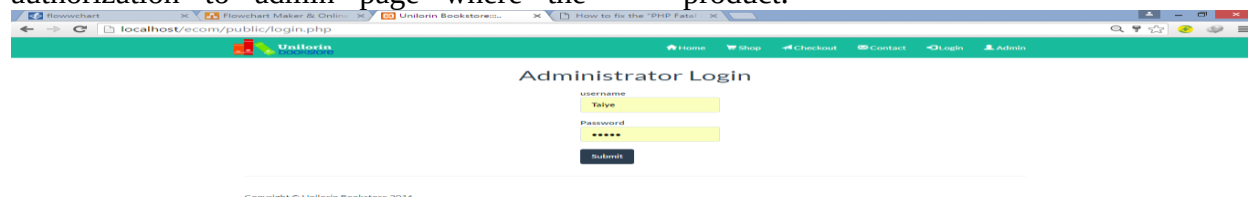


Figure 8: Admin login page.

Report page

This is a page on the administrator page where report about order can be seen, details

such as the product ID, product price, order ID, product title, and product quantity.

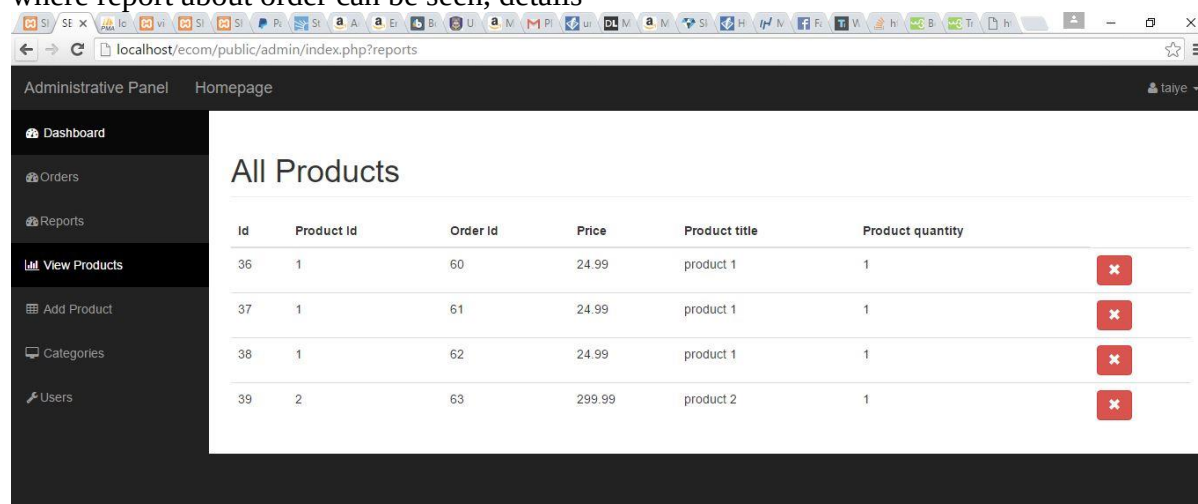


Figure 9: Report page

Order page

The order menu show case orders that have been completed, it show the amount,

transaction, ID, currency and status of the order.

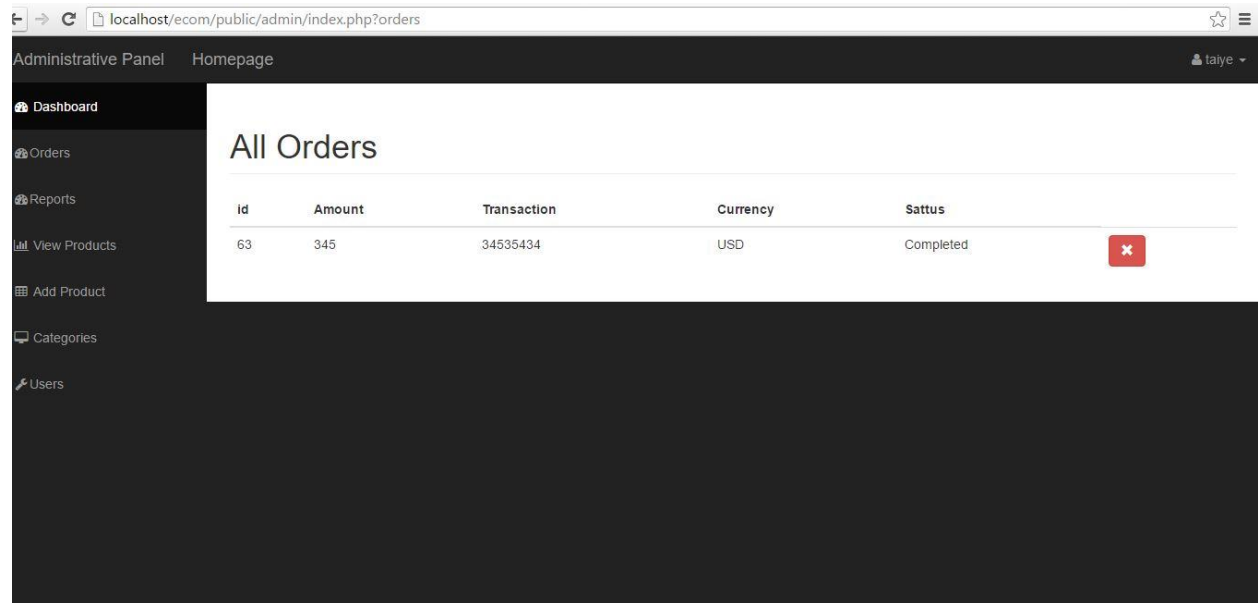


Fig 10: Order page

View product

This page showcase the product and the corresponding description, such as the ID,

Title category, price and quantity of the product with an option to delete a product.

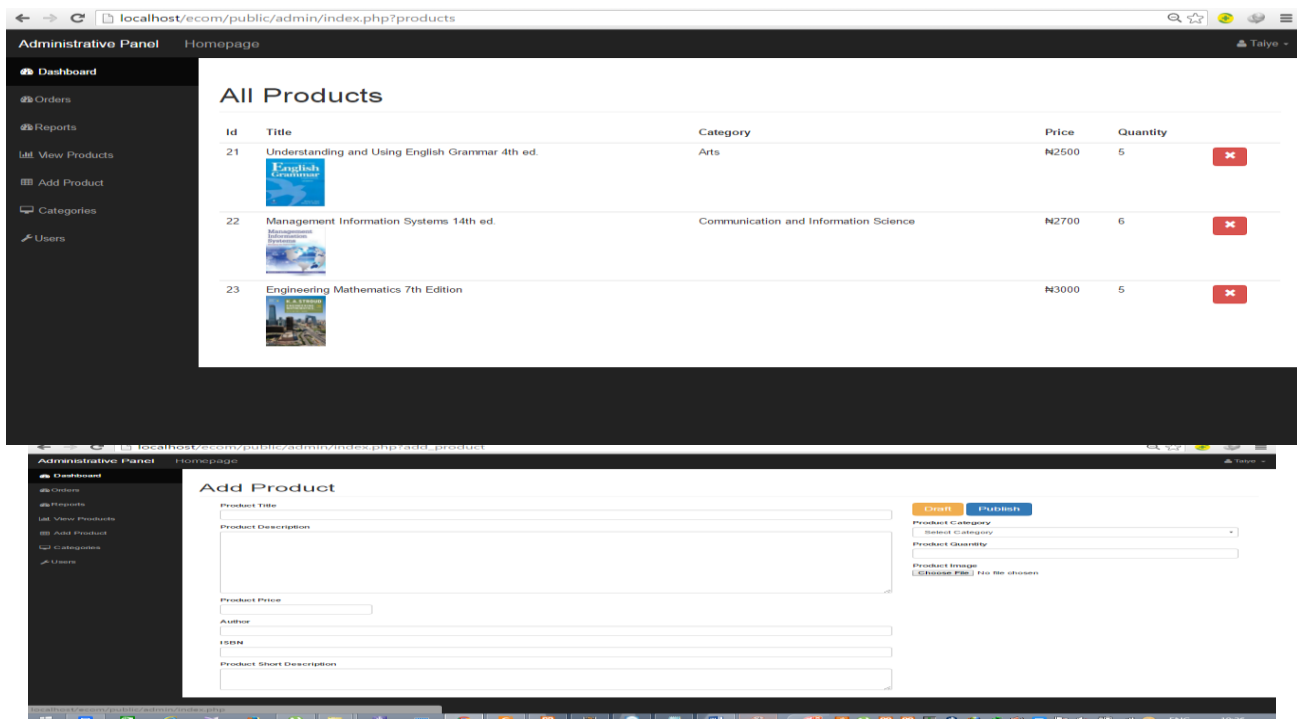


Figure 11: Add product page.

Categories page

Categories page shows different categories of product in the administrative area.

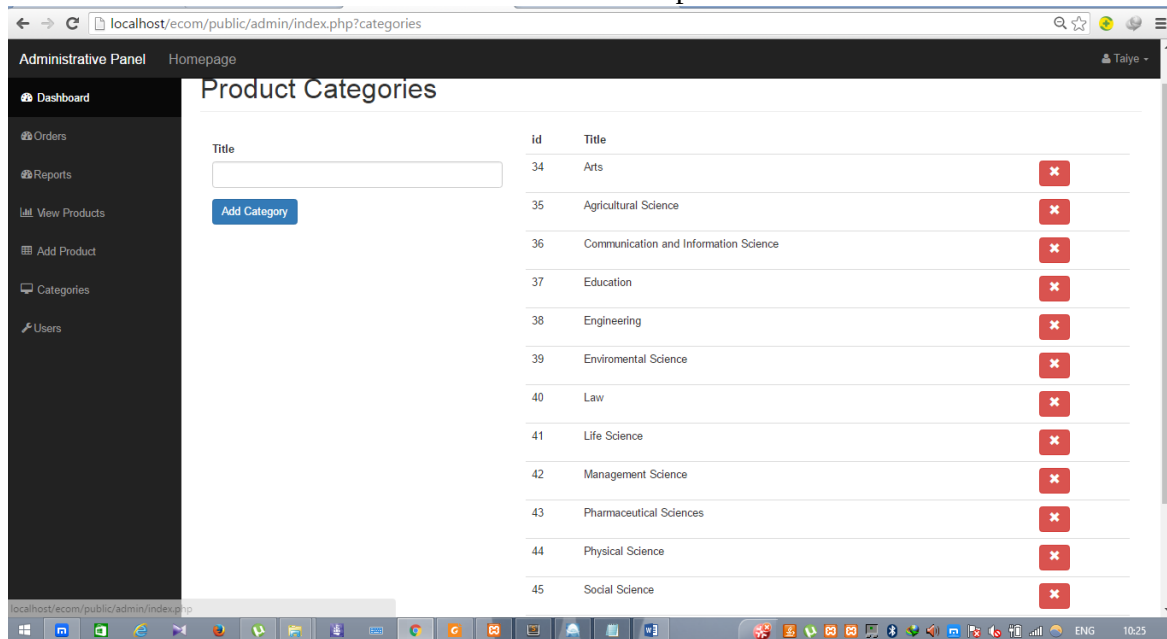


Figure 12: Categories

Users in admin

Users page the name and other details of registered user such as the user email, username and user password.

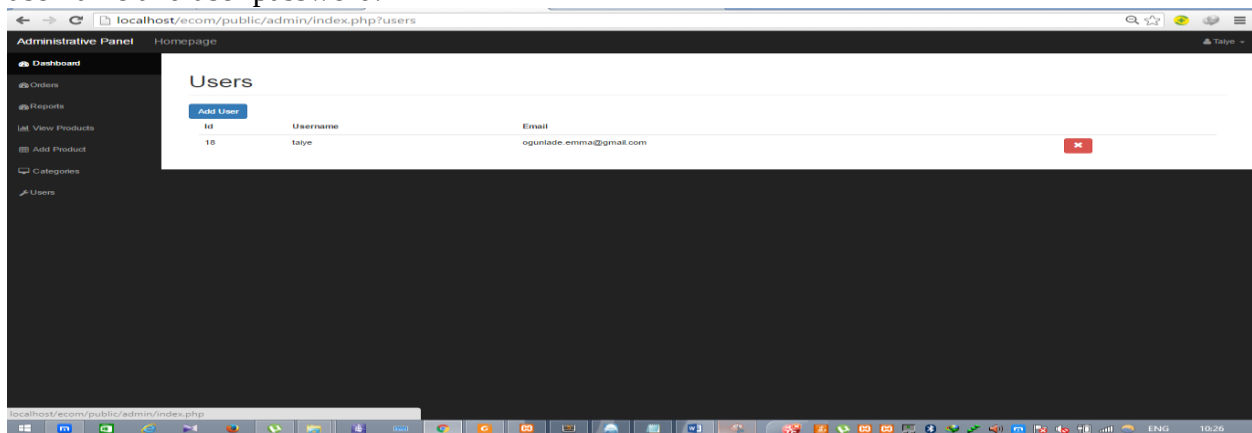


Figure 13: Users

Shop page

This page showcase different types of product uploaded irrespective of the

category, it also shows the search button which can be used to search for product.

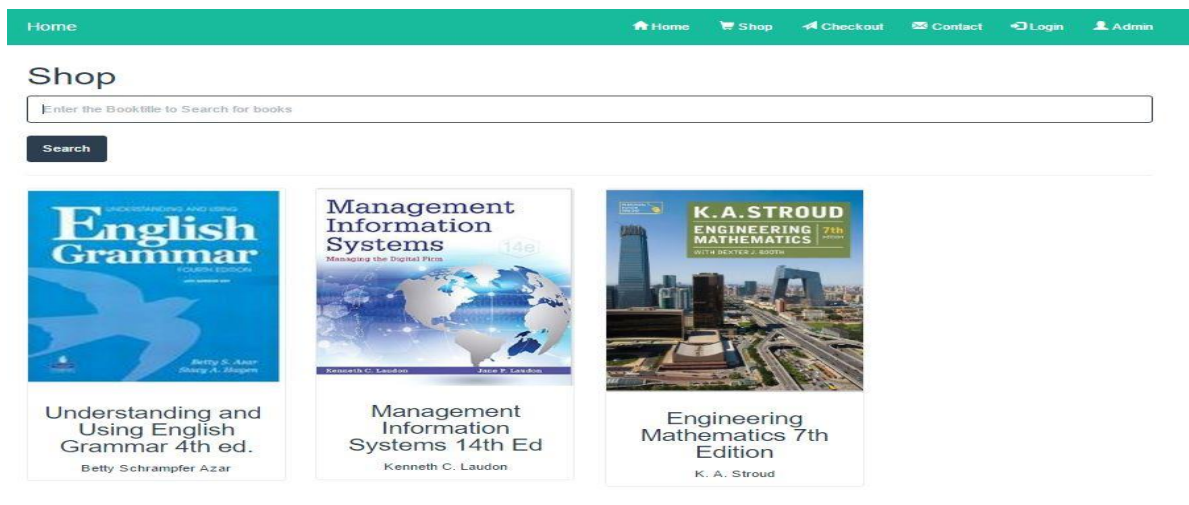


Figure 14: Shop page

Checkout page is a page that shows the product, price, quantity and cumulative price

of product based on quantity of the item selected. It also includes the subtotal

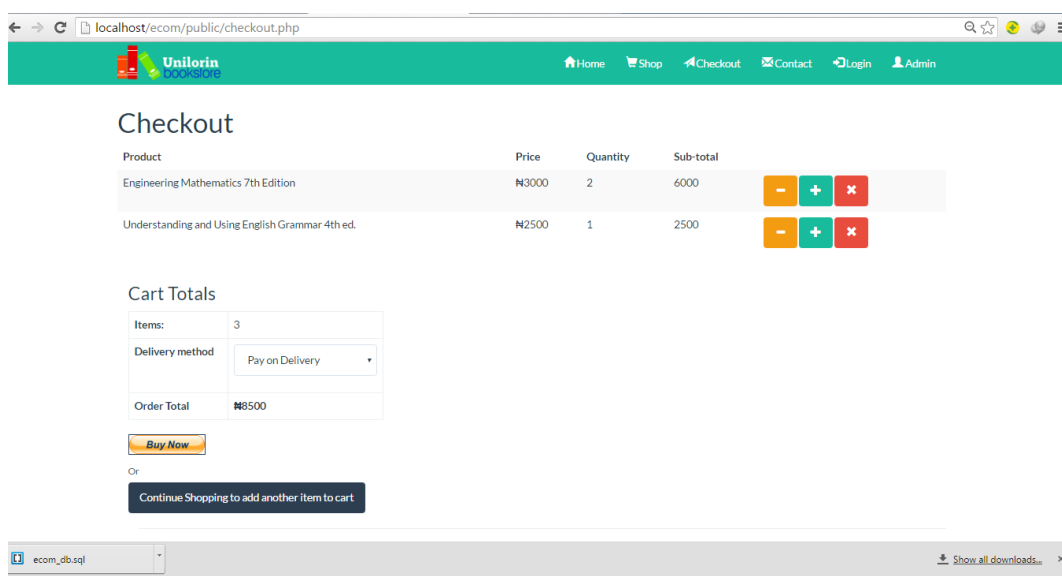


Figure 15: Check out page

Summary

The main aim of the development of this project was to transform UNILORIN bookstore system from its traditional way of purchasing books to using an online platform which operates electronically,

through the design and implementation of an online bookstore.

In achieving the necessary objectives for developing the new system, the existing system was studied, analyzed and corrected

to provide an efficiency and reliability system.

The main objective of the new system is to digitize all existing records of books kept by UNILORIN Bookstore and enabling some manual processes of the existing system to be done electronically. These processes include creating product categories, adding a new user admin, adding new products, modification of existing product details, getting order and report details.

Conclusion

This project has been able to fulfil its objective by developing an online bookstore using UNILORIN bookstore as the case study, it will serve the users and also ease the way they buy product online. It will facilitate 24/7 service to access the platform online and also provide a way of making comparison and selection of delivery method.

The system will be effective if it is used for the purpose it was meant for and other purposes that it can serve such as the sales of other learning materials available at the bookshop. Also, the awareness of this platform should be made known to students and staff of the university on the University's website, as it will help them get help in terms of how they can acquire academic materials and stationery materials which can help them in study or research purpose.

In addition to this, anyone who desires to work further on the same project can use another web stack such as Nodejs, MongoDB or Ruby on Rails or design a mobile app using hybrid technology.

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